

A Deployable Emergency Communications and Incident Management Platform for McHenry County

Executive Briefing for MCESV / MCEMA Leadership

THE PROBLEM WE SOLVE

When a disaster takes down commercial infrastructure — cellular networks, internet, power — the tools modern emergency management depends on disappear precisely when they are needed most. Incident logs revert to paper. Resource tracking becomes a whiteboard. ICS forms are filled out by hand, photocopied, and hand-carried between rooms. Situational awareness degrades to whatever one person can hold in their head. Every county that has worked a major activation knows this failure mode, and most have simply accepted it as the cost of doing business.

FieldComms eliminates that failure mode. It is a self-contained incident management platform that carries its own network, its own server, and its own tools — and it works with no internet, no cellular service, and no outside infrastructure of any kind.

WHAT FIELDCOMMS IS

FieldComms is a complete emergency operations platform built on two Raspberry Pi computers, three mesh Wi-Fi routers, and dual independent internet uplinks — all of which fits in a transport case and deploys in under 30 minutes. When it powers on, it broadcasts its own private Wi-Fi network called EMCOMM-NET. Any smartphone, tablet, or laptop within range connects and immediately has access to 32 professional emergency management tools through a standard web browser. No app installs. No accounts. No per-device configuration. No internet required for any core function.

CAPABILITY AREA	WHAT OPERATORS GET
Incident Command (ICS)	All five ICS sections, live T-card resource board, IAP development with the Planning P cycle, standard ICS forms with print output. Every position sees the same live data.
Radio Net Operations	Net loggers for amateur (ARES/RACES) and Starcom nets. Offline FCC lookup of 800,000 licensees. Automatic ICS-309 logs. Net inactivity alarm for operator safety.
Situational Awareness	Offline tactical maps with live APRS tracking, resource mapping, animated NWS radar, live weather alerts, HF propagation.
Resource & Personnel	Member roster with certifications and check-in, facility directory, resource board, printable access cards for rapid mutual-aid onboarding.
Communications	Winlink radio email, JS8Call HF messaging, APRS, and AMPRNet — data paths that function when commercial networks do not.
Reference & Printing	Offline Wikipedia and medical references, county SOGs, radio cheat sheets, shared print center producing complete IAP packages on site.

BUILT FOR THE WORST DAY

Every layer has a backup, and complete loss of outside infrastructure still leaves a fully functioning EOC. Internet comes first from dual-carrier cellular (T-Mobile and Verizon), with automatic failover to Starlink satellite

within 60 seconds. If both fail, all 32 tools continue on the local network — only live weather feeds pause. Storage is mirrored so a drive failure costs nothing. The Wi-Fi mesh covers a multi-room EOC or outdoor base camp with seamless roaming, and the entire system runs on battery or generator power.

WHERE IT WORKS

SCENARIO	HOW FIELDCOMMS PERFORMS
All-hazards EOC activation	Each ICS section works in its own room, all synchronized live. IAP developed and printed on site. Amateur and Starcom nets logged simultaneously.
Search and rescue	Parking-lot base camp. Mutual-aid crews check in against the roster in seconds. Sectors briefed from a shared map, teams tracked on the tactical display, storms watched on radar.
Weekly readiness nets	The same loggers run the routine weekly nets — every operator knows the tools before an activation, and every net produces an archivable professional log.
Planned events	Parades, marathons, fairs — resource tracking and documentation that builds operator proficiency year-round.

WHAT IT COSTS — AND WHAT IT REPLACES

Per the project bill of materials at verified mid-2026 pricing, the core FieldComms infrastructure — both servers with mirrored storage, three mesh routers, 16-port switch, InstyConnect cellular and Starlink satellite uplinks, four operator workstations with monitors, GPS, backup drives, and a color laser printer — totals approximately **\$6,051**. The complete radio station (IC-7300MK2 transceiver, Astron power supply, LDG station tuner, Chameleon MPAS 2.0 field antenna system with URT1 remote tuner and 300 ft of antenna wire, and laptop — items many clubs already partially own) adds roughly **\$3,900**, bringing a complete from-scratch build to approximately **\$14,075**. Note that current Raspberry Pi and SSD pricing reflects the industry-wide 2026 memory shortage and is expected to decline. Recurring costs are limited to cellular data (\$5/month standby, \$79-99/month only during activations) and Starlink service as the county elects. There are no software licenses, no per-seat fees, and no subscriptions — the software is built on open-source foundations and owned outright by MCESV.

Commercial incident management platforms with comparable capability are sold as annual subscriptions commonly running \$10,000 to \$50,000 per year — and every one of them assumes working internet. FieldComms delivers the capability class of those systems, works when they cannot, and costs less in total than a single year of their licensing.

WHY THIS MATTERS TO LEADERSHIP

Continuity of operations. The county gains an EOC capability that survives total infrastructure loss — the exact condition that triggers an EOC activation in the first place.

Professional documentation. Every net, every incident, every resource movement is logged digitally in FEMA-standard ICS formats — strengthening after-action review, reimbursement claims, and legal defensibility.

Mutual aid readiness. Any responder with a phone can be onboarded in under a minute. Nothing to install, nothing to configure — connect to Wi-Fi and work.

Volunteer force multiplication. The system was designed and built entirely by MCESV volunteers, is maintained by volunteers, and is fully documented in over 190 pages of professional manuals — installation, operations, field quick-reference, and a beta test program. Institutional knowledge lives in the documents, not

one person.

Grant positioning. As a 501(c)(3)-operated emergency communications asset, FieldComms aligns with FEMA BRIC, DHS Nonprofit Security Grant Program, ARDC, and ARRL Foundation funding priorities — the documentation package required by grant applications already exists.

CURRENT STATUS AND THE ASK

FieldComms v1.0 is feature-complete, fully documented, and entering field beta testing with a structured 22-page test protocol. The software is version-controlled, backed up, and reproducible. We are requesting leadership endorsement to: (1) complete beta testing in exercise conditions; (2) present the system at a county emergency management exercise for evaluation; and (3) incorporate FieldComms into MCESV standard operating guidelines as the primary incident documentation platform for activations and nets.

Prepared by Jim Rospopo, KE4CON · McHenry County Emergency Services Volunteers · K9ESV · Live demonstration available on request — the system deploys in 30 minutes.